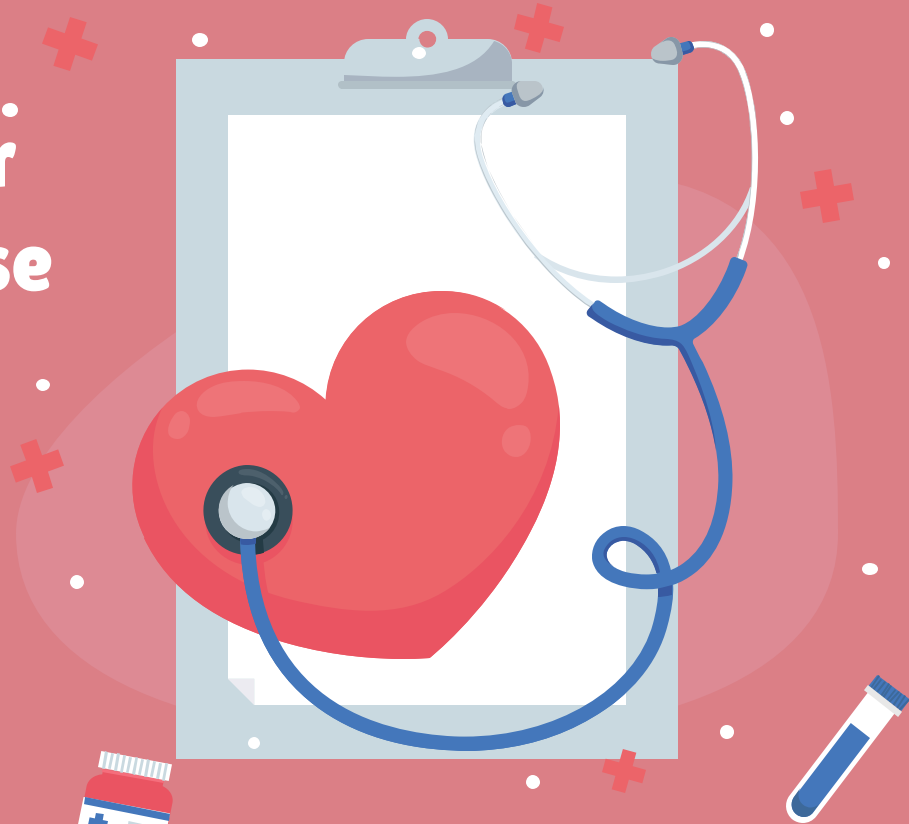
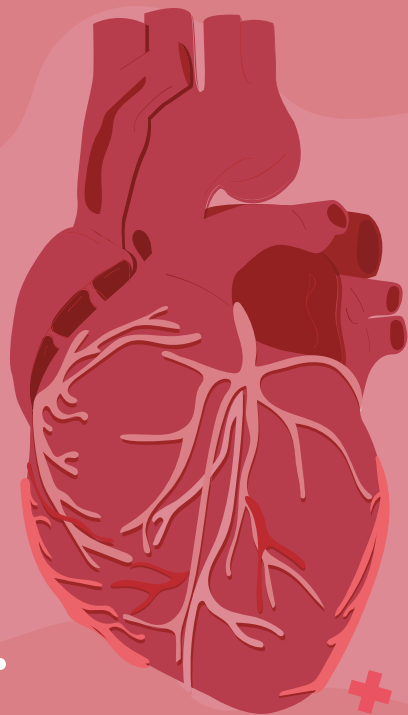


Machine Learning for Cardiovascular Disease Detection and Risk Stratification

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ENBC 321





INTRODUCTION

- CVD is the world's leading cause of death, claiming 20.5 million lives annually with 75% of fatalities occurring in low- and middle-income countries.
- While heart attacks and strokes cause 85% of deaths, the disease encompasses various conditions affecting heart rhythm, valves, and blood vessels.
- Presentation varies by sex, as women frequently experience atypical signs like extreme fatigue and nausea rather than traditional chest pain.
- Machine learning optimizes early intervention by analyzing biometric patterns to predict individual cardiac events before they occur.



Research Question:

**Which clinical features best predict CVD
outcomes, and which ML models
perform best?**

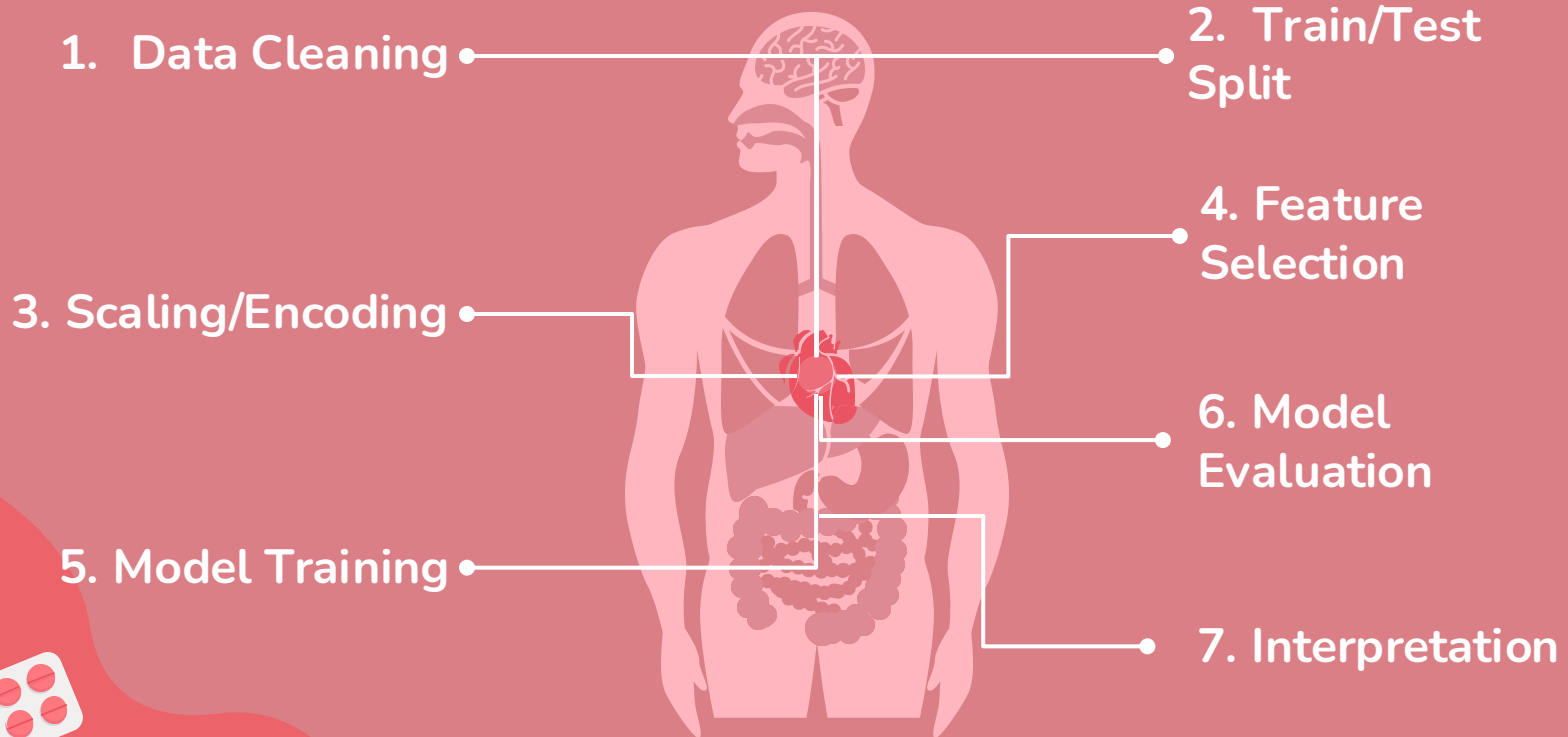


Dataset Overview



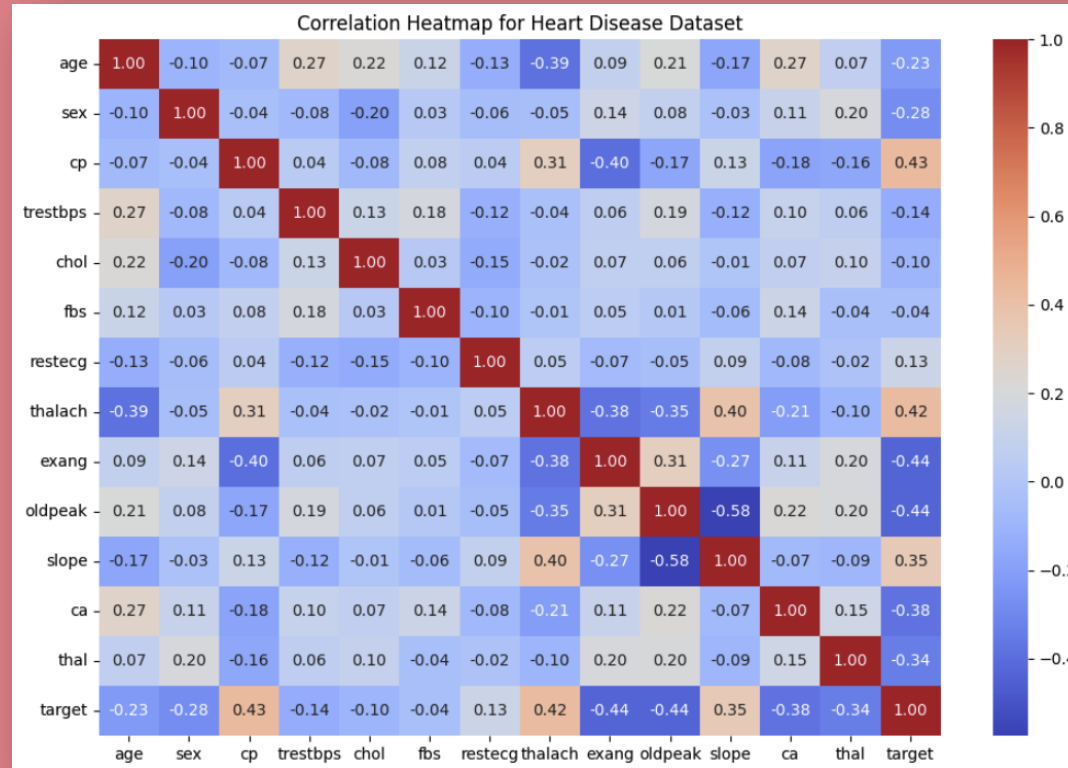
Dataset	Goal	Target	# Samples	# Total Features	ML Task	Main Features
Dataset 1: Heart Disease	Predict disease presence	target	1025	13	Binary classification	age, sex, chest pain, cholesterol, max heart rate, oldpeak
Dataset 2: CVD Risk	Predict risk level	CVD Risk Level	1529	22	Multiclass classification	BMI, blood pressure, cholesterol, diabetes, physical activity

Methods / ML Workflow

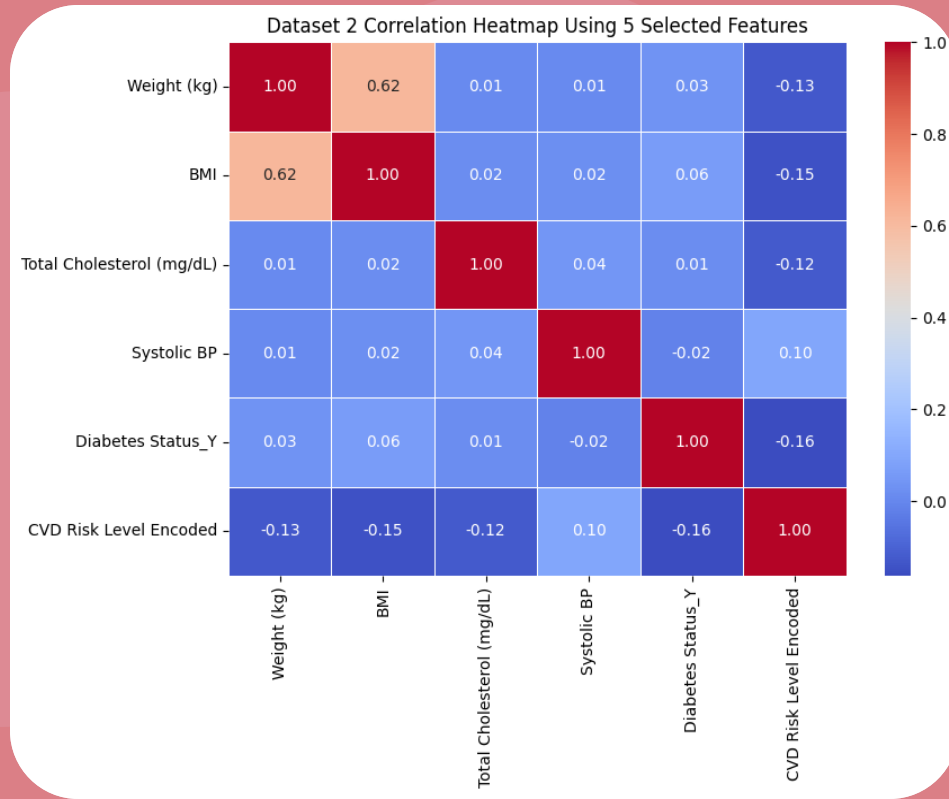


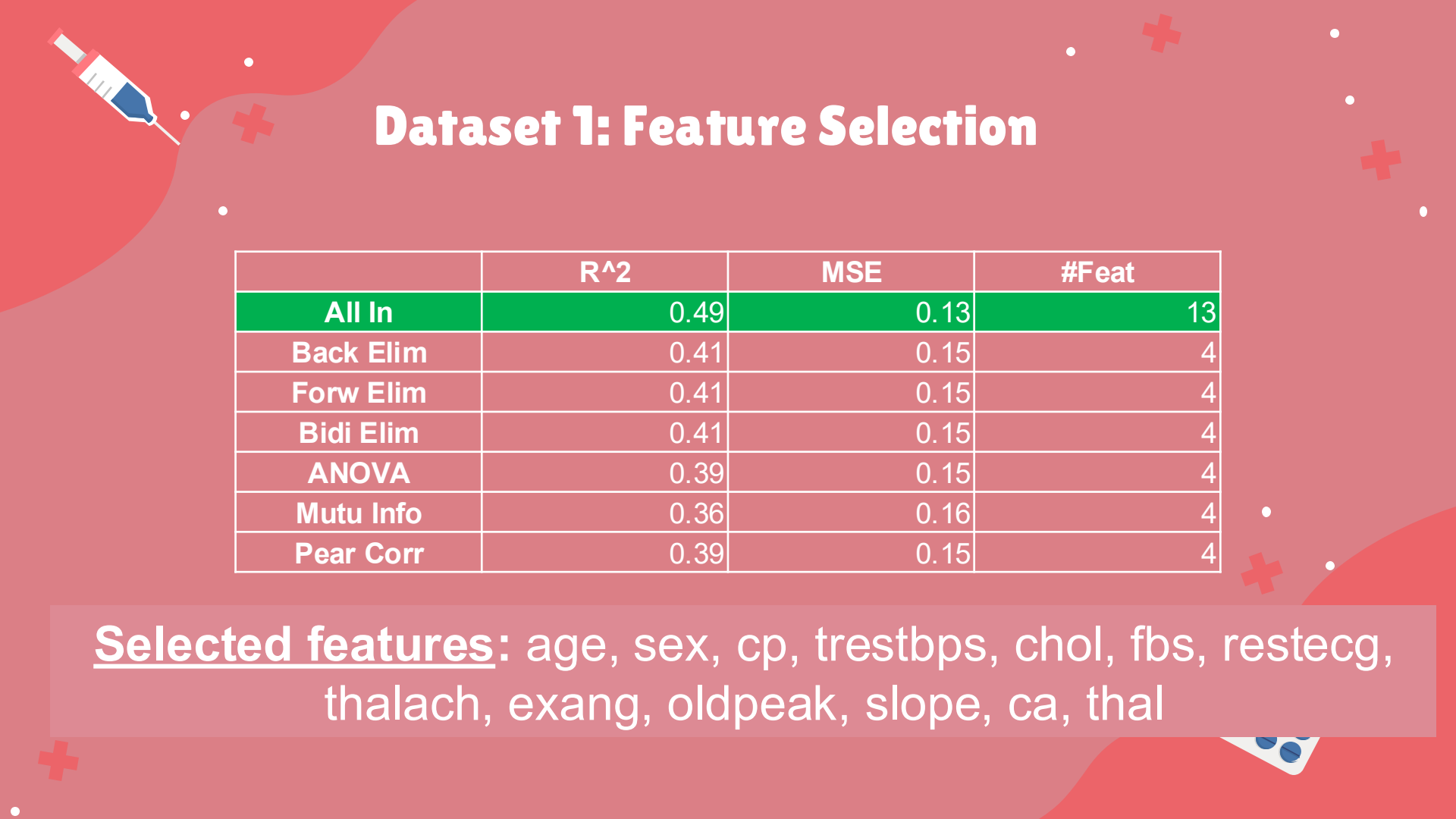
PREV NEXT

Dataset 1: Initial Feature Analysis



Dataset 2: Initial Feature Analysis

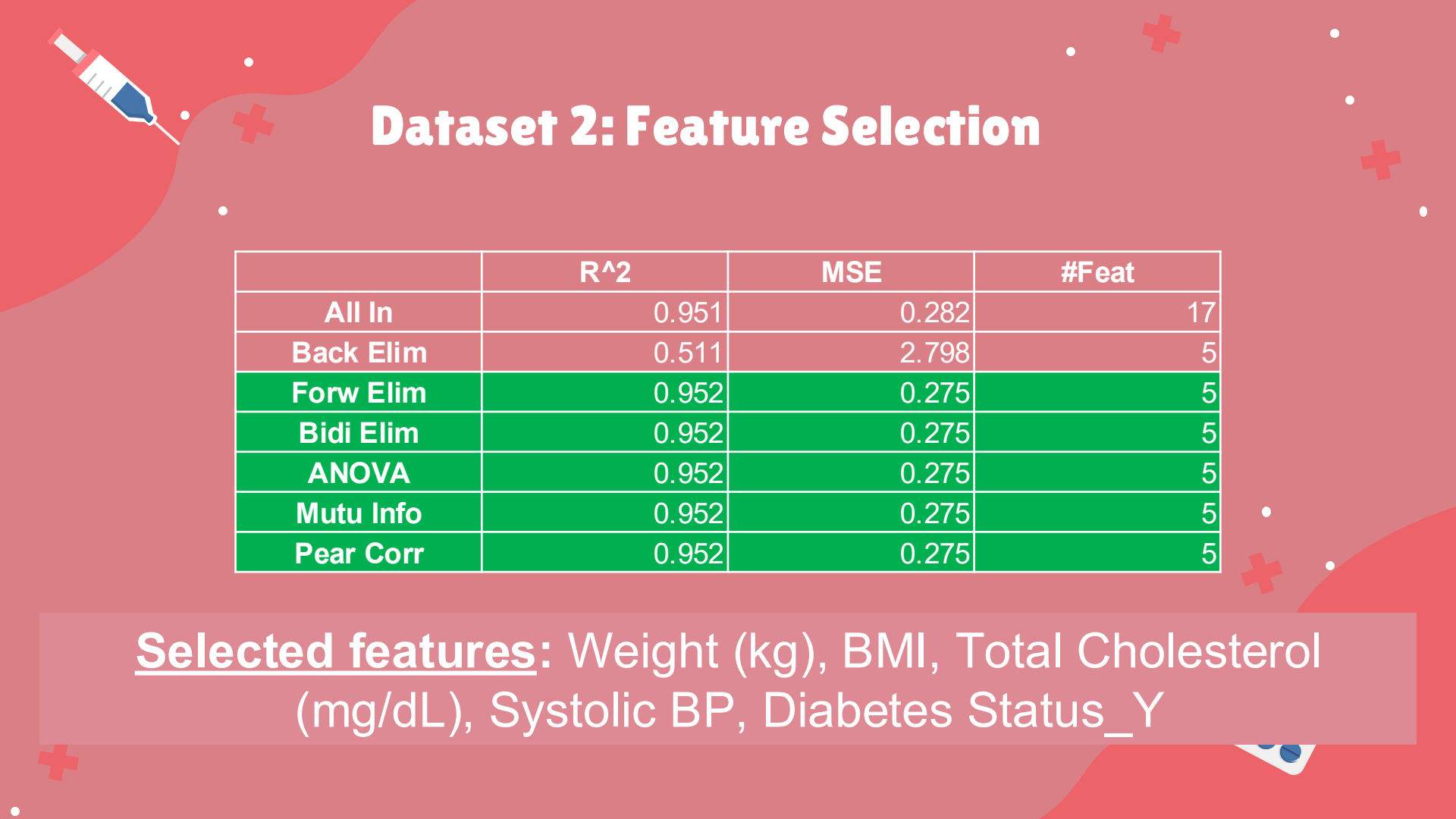




Dataset 1: Feature Selection

	R^2	MSE	#Feat
All In	0.49	0.13	13
Back Elim	0.41	0.15	4
Forw Elim	0.41	0.15	4
Bidi Elim	0.41	0.15	4
ANOVA	0.39	0.15	4
Mutu Info	0.36	0.16	4
Pear Corr	0.39	0.15	4

Selected features: age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal



Dataset 2: Feature Selection

	R^2	MSE	#Feat
All In	0.951	0.282	17
Back Elim	0.511	2.798	5
Forw Elim	0.952	0.275	5
Bidi Elim	0.952	0.275	5
ANOVA	0.952	0.275	5
Mutu Info	0.952	0.275	5
Pear Corr	0.952	0.275	5

Selected features: Weight (kg), BMI, Total Cholesterol (mg/dL), Systolic BP, Diabetes Status_Y

Train Test/Split



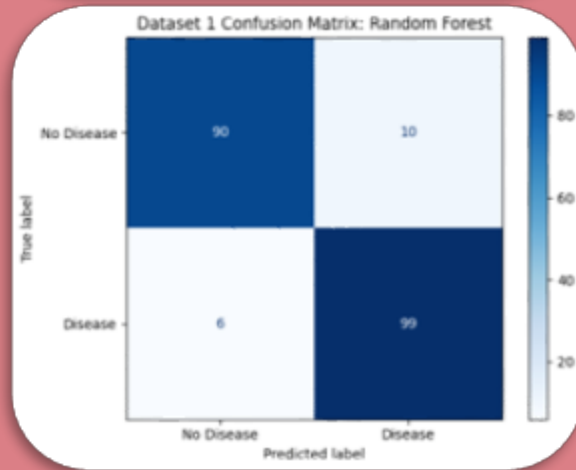
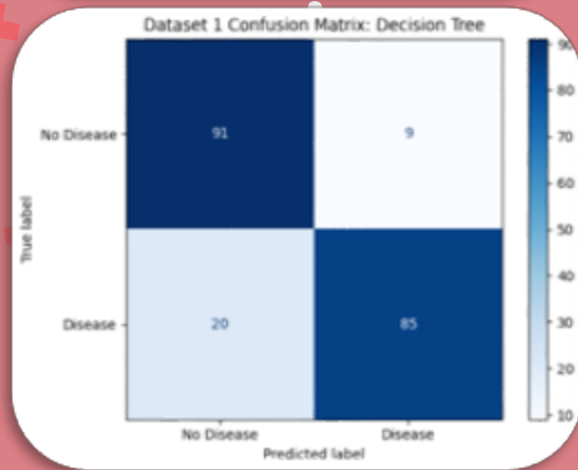
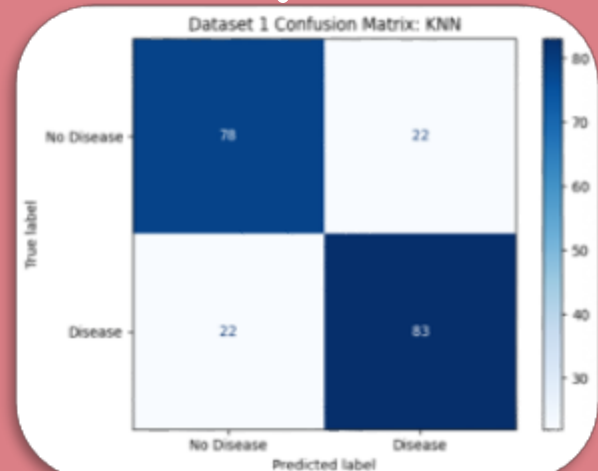
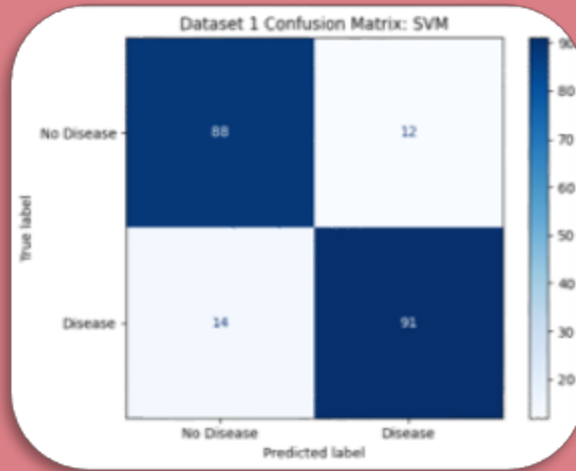
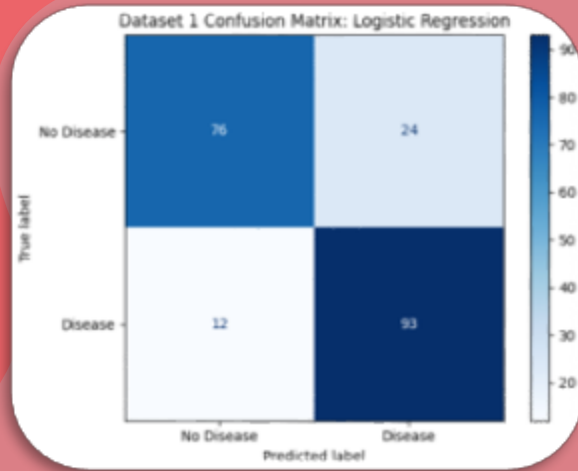
Splitting

The dataset is partitioned into an 80% training set to facilitate robust model learning and a 20% testing set for unbiased performance evaluation.

Scaling

Scikit-learn's StandardScaler is applied to both datasets to normalize feature variance and ensure model convergence while strictly preventing data leakage.

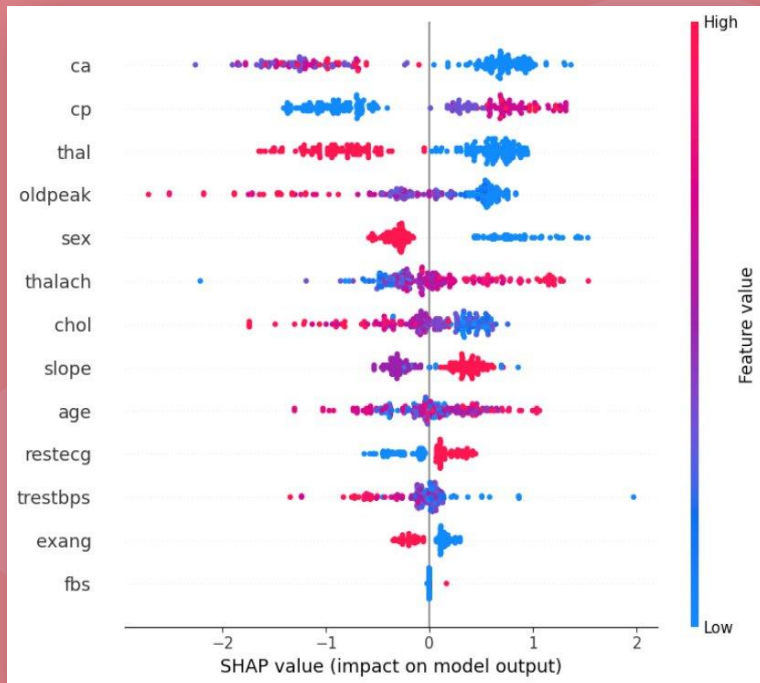
Dataset 1: Model Classification Performance



Dataset 1: Model Evaluation

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Gradient Boosting	0.94	0.93	0.96	0.94	0.99
Random Forest	0.92	0.91	0.94	0.93	0.97
SVM	0.87	0.88	0.87	0.88	0.96
Decision Tree	0.86	0.90	0.81	0.85	0.94
KNN	0.79	0.79	0.79	0.79	0.93
Logistic Regression	0.82	0.80	0.89	0.84	0.88

Dataset 1: Best Model Feature Importance

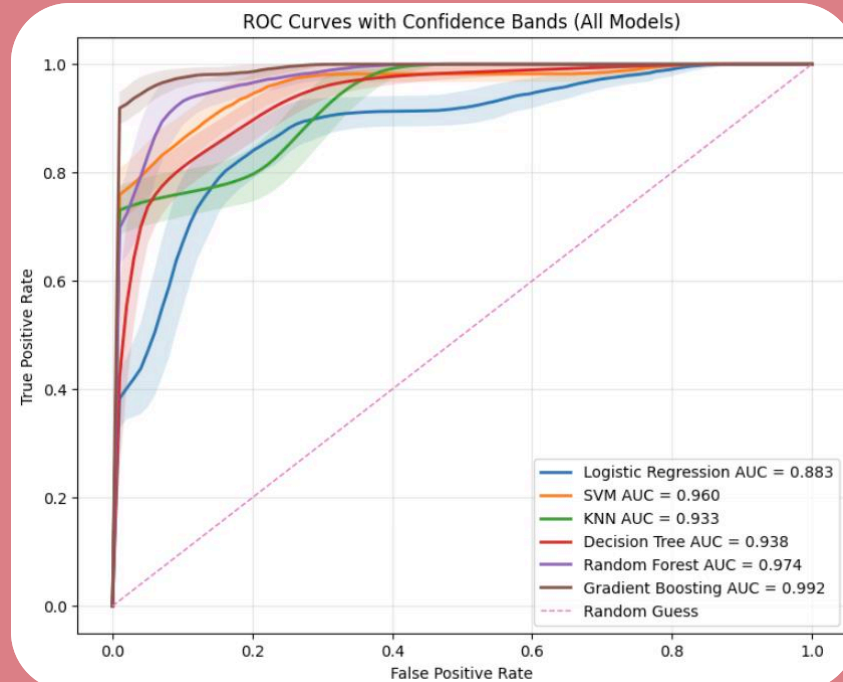


Blue shows low feature value importance, while red indicates high importance.

Horizontal positioning shows positive or negative impact

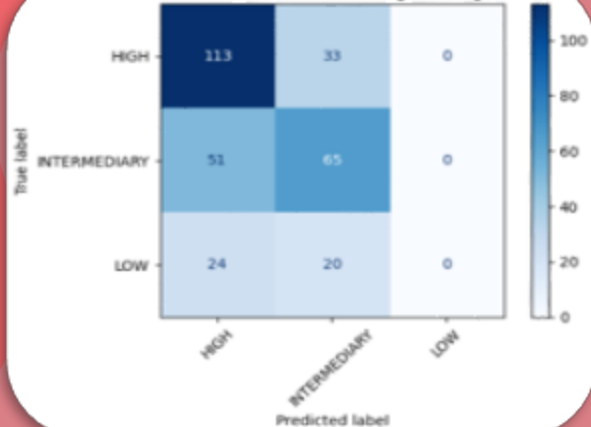
Gradient Boosting feature importance shown through SHAP plot.

Dataset 1: Model Evaluation

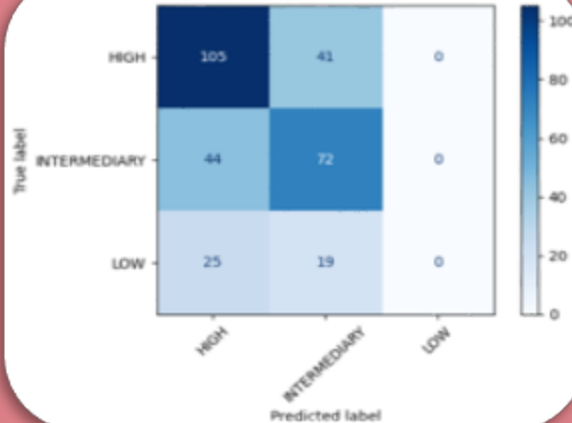


Dataset 2: Model Classification Performance

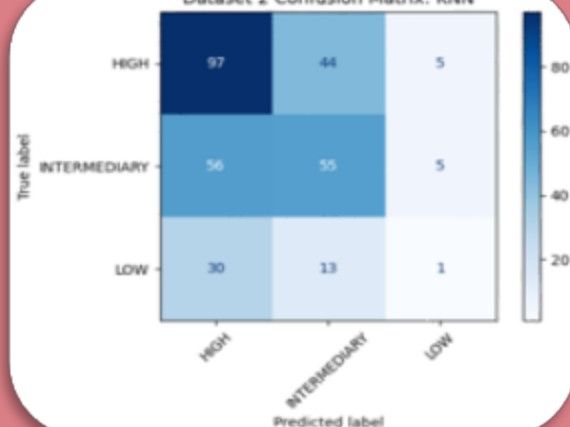
Dataset 2 Confusion Matrix: Logistic Regression



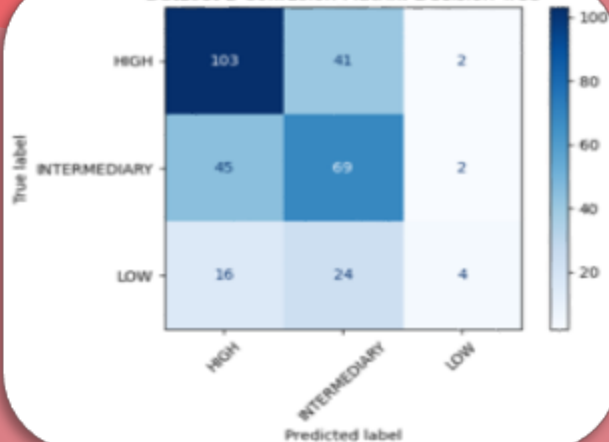
Dataset 2 Confusion Matrix: SVM



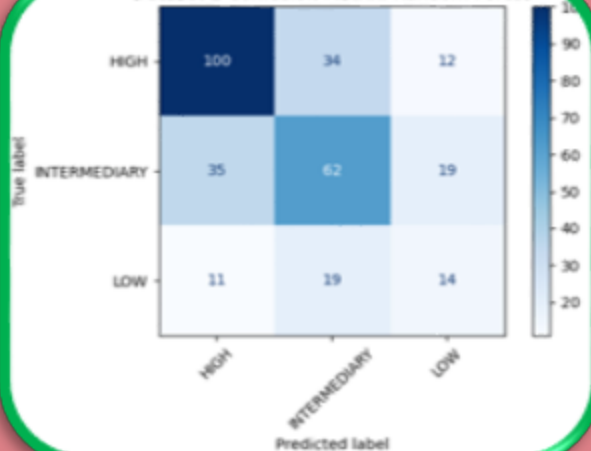
Dataset 2 Confusion Matrix: KNN



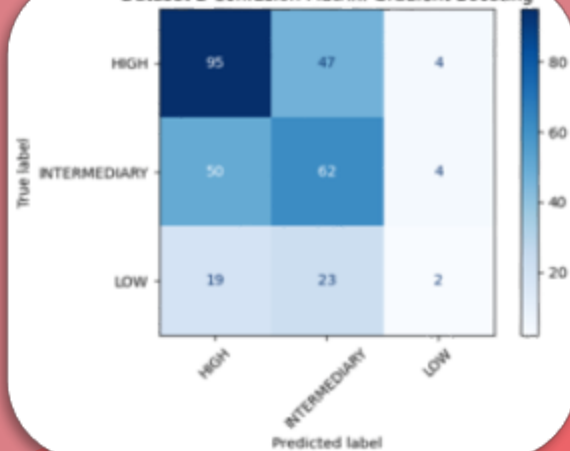
Dataset 2 Confusion Matrix: Decision Tree



Dataset 2 Confusion Matrix: Random Forest



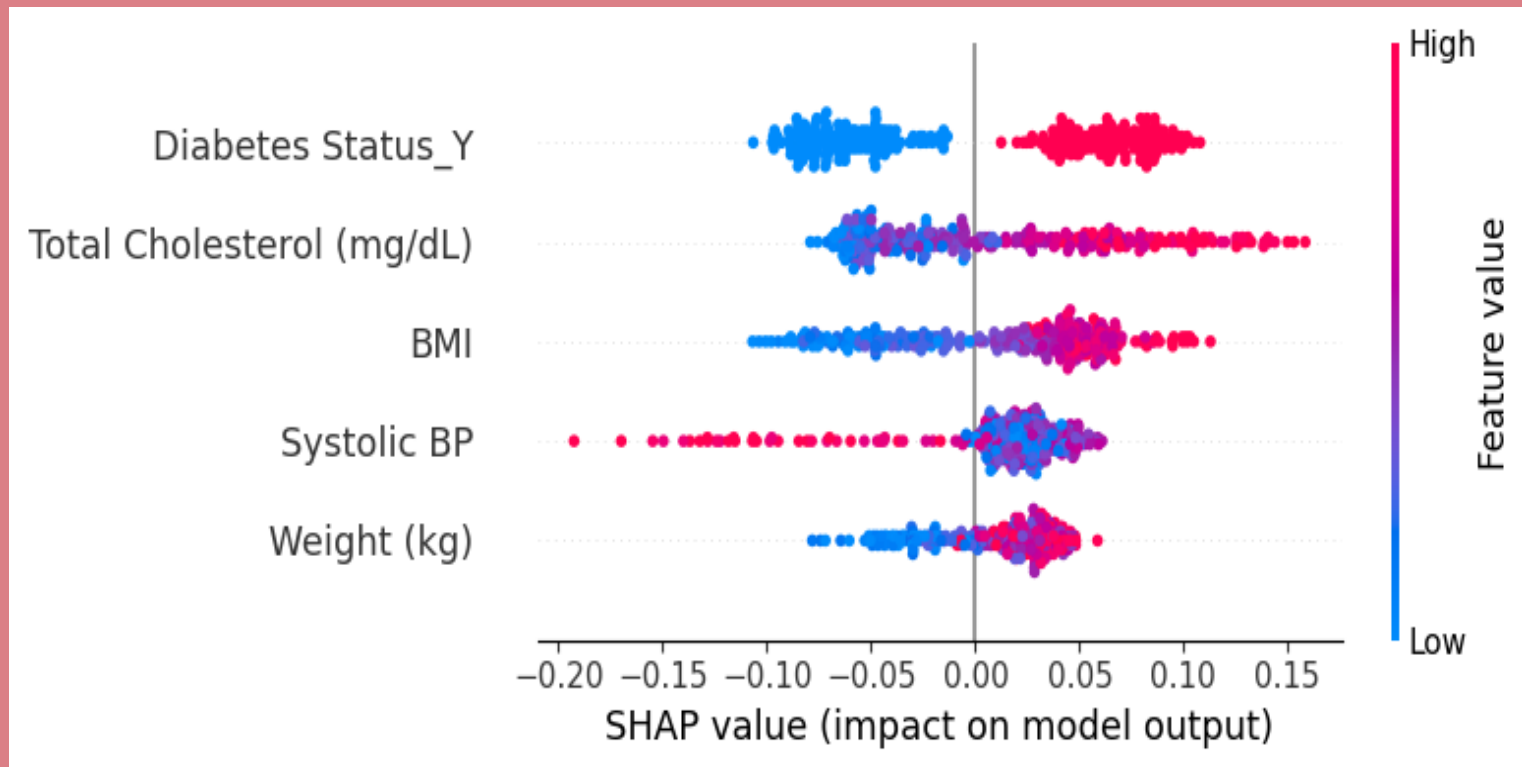
Dataset 2 Confusion Matrix: Gradient Boosting



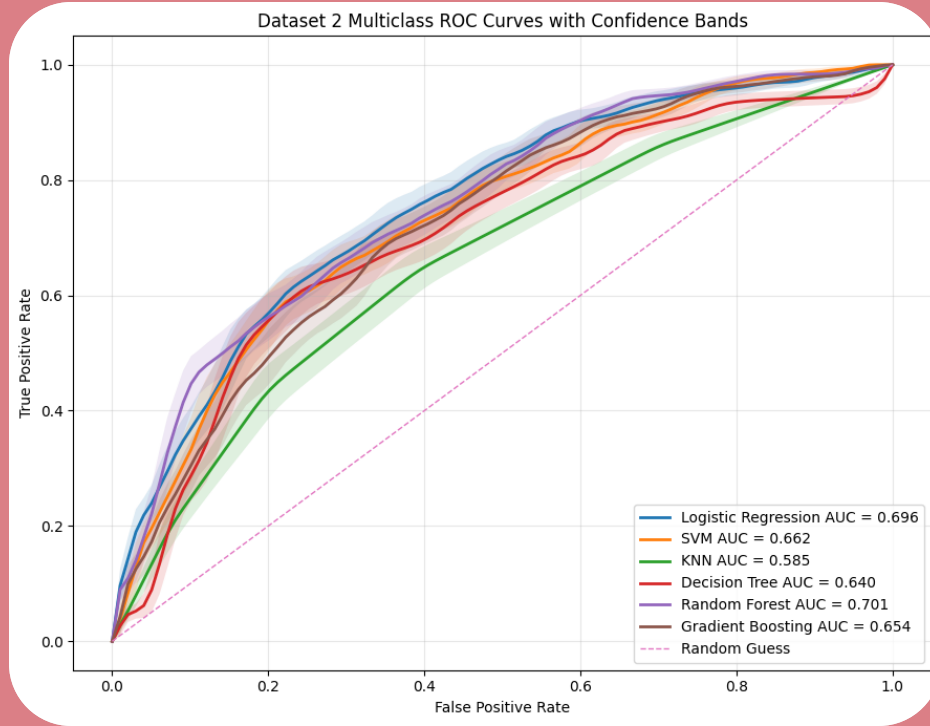
Dataset 2: Model Evaluation

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Random Forest	0.552	0.480	0.552	0.513	0.697
SVM	0.578	0.494	0.578	0.533	0.662
Gradient Boosting	0.519	0.483	0.519	0.492	0.653
Logistic Regression	0.581	0.495	0.581	0.533	0.696
Decision Tree	0.575	0.566	0.575	0.548	0.640
KNN	0.500	0.452	0.500	0.469	0.585

Dataset 2: Initial Feature Importance (High Risk Class)



Dataset 2: Model Evaluation Cont.



Limitations

- **Dataset 1 was still limited in scale:** Although Dataset 1 had **1,025 samples**, it is still relatively small for building a highly generalizable clinical ML model.
- **Dataset 2 had weaker class separation:** The multiclass models often confused neighboring risk groups, especially intermediate-risk patients.
- **Limited clinical feature depth:** The datasets did not include all potentially important factors, such as medication use, family history, imaging data, longitudinal follow-up, or detailed lifestyle history.
- **Generalizability is uncertain:** Both datasets should be externally validated on larger and more diverse patient populations before any real clinical use.

Conclusions

- **Binary CVD detection performed strongly:** Dataset 1 achieved high predictive performance, with Gradient Boosting reaching **94% accuracy** and **0.99 ROC-AUC**.
- **Dataset 1 key predictors:** SHAP analysis showed that **ca, chest pain type, thalassemia, oldpeak, maximum heart rate, and sex** contributed strongly to disease prediction.
- **Multiclass risk stratification was more difficult:** Dataset 2 models showed moderate performance, with the best accuracy around **58%** and ROC-AUC around **0.70**, indicating that low, intermediate, and high risk groups were harder to separate.
- **Selected Dataset 2 features were clinically relevant:** The strongest selected variables included **weight, BMI, total cholesterol, systolic blood pressure, and diabetes status**.



Future Work

- **Expand and diversify the datasets:** Use larger patient cohorts across different hospitals, age groups, sex groups, and populations.
- **Improve multiclass risk prediction:** Test stronger hyperparameter tuning, class balancing, and additional clinical variables to better separate low, intermediate, and high-risk patients.
- **Perform external validation:** Evaluate the models on independent CVD datasets to test whether performance holds outside the training data.
- **Explore demographic-aware modeling:** Analyze model performance by sex, age, and other demographic variables to check for bias or performance differences.

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THANK YOU



Questions?